

Task 3: Attention Visualization & Token Attribution

TinyLlama Instruction Tuning Project

March 24, 2026

Task description

Subtasks: (1) Load trained TinyLlama LoRA checkpoint. (2) Extract attention maps from all heads for 20 sample instructions. (3) Cluster attention heads by cosine similarity. (4) Token attribution via attention rollout. (5) HTML heatmaps for 5 detailed + 20 total generations. (6) Identify *instruction-following* vs *completion* heads empirically. (7) Compare base model vs LoRA-tuned model quantitatively.

Model: TinyLlama-1.1B-Chat-v1.0 + LoRA (adapters/tinyllama-lora-alpaca, rank 16, $\alpha=32$); **704 total heads** (22 layers $\times$ 32 heads); 20 Alpaca prompts; script: `task3_attention_visualization.py`.

Procedure & technical implementation

- Attention extraction.** Monkey-patch `self_attn._call_` to capture softmax weights (H, L, L) per layer in a post-generation full-sequence forward pass (KV-cache calls skipped so only one hook fires per input).
- Attention rollout.** Cumulative product $R^{(l)} = \text{norm}(A^{(l)} + I) \cdot R^{(l-1)}$ across all 22 layers; output is one attribution vector per sample ranking input tokens by influence on the last output position.
- Head clustering.** Feature vector per head = concatenated mean instruction- and completion-attention over 20 samples (dim = 2×20); cosine distance + average-linkage hierarchical clustering (`scipy`), $k=3$ groups, applied independently to base and LoRA.
- Head classification.** Per head: if mean response $\rightarrow$ prompt attention $\geq$ mean local-window response attention $\Rightarrow$ **instruction-following**; else **completion**. Threshold applied across all 20 prompts.
- Base vs LoRA comparison.** Layer-wise $\Delta_{\text{net}} = (\text{LoRA instr} - \text{LoRA comp}) - (\text{Base instr} - \text{Base comp})$; heatmap and scatter over all 704 heads. Outputs: `results/task3/` — 20 HTML heatmaps, 3 clustering PNGs, `analysis_summary.json`, `layer_analysis.json`, `head_classification.json`.

Results

Table 1: Head classification and attention statistics (20 Alpaca prompts, 704 heads). Source: `analysis_summary.json`.

Model	Instruction-following	Completion	Attn. focus	Entropy
Base	445 (63.2%)	259 (36.8%)	0.5445	0.7355
LoRA	417 (59.2%)	287 (40.8%)	0.5999	0.7289
Δ	-28 heads (-4.0%)	+28	+10.2%	-0.9%

Key findings.

- 28 heads** (4%) shifted instruction $\rightarrow$ completion after LoRA; all 22 layer Δ_{net} values are negative, confirming a uniform completion-mode amplification across depth.
- Attention focus +10.2%** (0.5445 $\rightarrow$ 0.5999) with slightly lower entropy: LoRA sharpens prompt-token attribution without dispersing attention.
- Most affected layers:** 16 ($\Delta=-0.00301$), 12 (-0.00248), 13 (-0.00243), 9 (-0.00236), 2 (-0.00208) — mid-to-late layers dominate LoRA’s recalibration.
- Stable instruction heads:** L8H11, L15H16, L11H8, L15H20 retain near-identical scores in both models; L1H30 and L14H30 completion heads are amplified in magnitude by LoRA.
- Cluster structure preserved:** 3-group dendrogram topology identical in base and LoRA; LoRA makes within-cluster refinements only.

Evidence (screenshots)

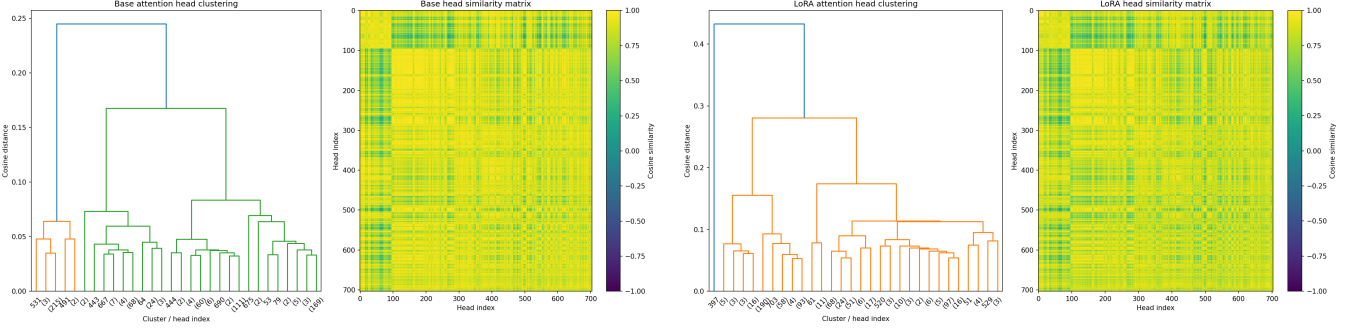


Figure 1: Attention head clustering dendrograms (left: base; right: LoRA-tuned). Cosine-distance average-linkage, $k=3$. Both models share the same 3-cluster topology; LoRA modestly shifts within-cluster similarity.

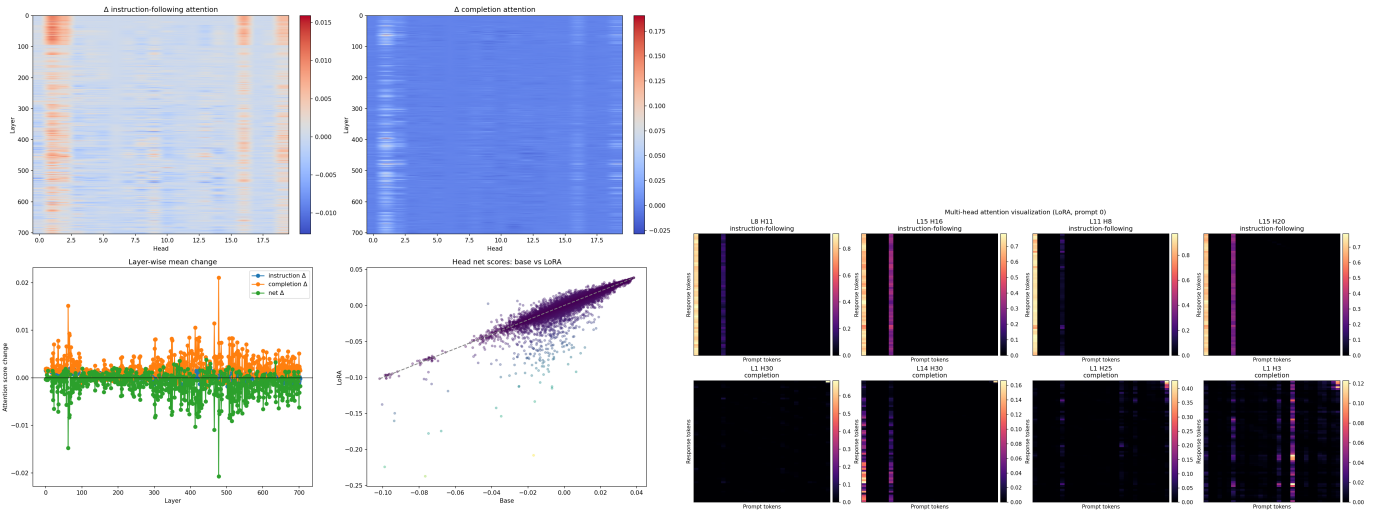


Figure 2: **Left:** Δ attention heatmaps (LoRA–Base) per (layer, head) for instruction and completion modes, layer-wise means, and per-head scatter. Most points lie below the diagonal, confirming the completion shift. **Right:** Multi-head attention matrices for prompt 0 (LoRA). Top row: 4 most instruction-following heads; bottom: 4 most completion-oriented heads.